Rule of Inference for checking the validity of an Argument

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First, we describe the process of derivation by which one can check the validity of an argument. In the process of derivation, three rules called *Rule P*, *Rule T*, and *Rule CP* are used, which are described as follows.

Rule P: A premise may be introduced at any point of derivation.

Rule T: A formula S may be introduced in a derivation if S is tautologically implied by any one or more of the preceding formulas in the derivation.

Rule CP: If we can derive s from r and a set of premises, then we can derive $r \to s$ from the set of premises alone.

The following table gives a list of rules of inference which are often used to check the validity of an argument.

No.	**Rule of Inference**	**Tautological Form**	**Identified by**
I_1	$\frac{p}{\therefore p \vee q}$ or $\frac{q}{\therefore p \vee q}$	$p \to p \vee q$ or $q \to p \vee q$	Addition
I_2	$\frac{p \wedge q}{\therefore p}$ or $\frac{q \wedge p}{\therefore q}$	$(p \wedge q) \to p$ or $(q \wedge p) \to q$	Simplification
I_3	$\frac{p,\ q}{\therefore p \wedge q}$ or $\frac{q,\ p}{\therefore q \wedge p}$	$((p) \wedge (q)) \to (p \wedge q)$ or $((q) \wedge (p)) \to (q \wedge p)$	Conjunction
I_4	$\frac{p \to q,\ p}{\therefore q}$	$((p \to q) \wedge p) \to q$	Modus Ponens
I_5	$\frac{p \to q,\ \sim q}{\therefore \sim p}$	$((p \to q) \wedge (\sim q)) \to \sim p$	Modus Tollens
I_6	$\frac{p \to q,\ q \to r}{\therefore p \to r}$	$((p \to q) \wedge (q \to r)) \to (p \to r)$	Hypothetical Syllogism
I_7	$\frac{p \vee q,\ \sim q}{\therefore p}$ or $\frac{p \vee q,\ \sim p}{\therefore q}$	$((p \vee q) \wedge \sim q) \to p$ or $((p \vee q) \wedge \sim p) \to q$	Disjunctive Syllogism
I_8	$\begin{array}{c} p \vee q \\ p \to r \\ q \to r \\ \hline \therefore r \end{array}$	$((p \vee q) \wedge (p \to r) \wedge (q \to r)) \to r$	Dilemma
I_9	$\begin{array}{c} p \vee r \\ p \to q \\ r \to s \\ \hline \therefore q \vee s \end{array}$	$((p \vee r) \wedge (p \to q) \wedge (r \to s)) \to (q \vee s)$	Constructive Dilemma
I_{10}	$\begin{array}{c} \sim q \vee \sim s \\ p \to q \\ r \to s \\ \hline \therefore \sim p \vee \sim r \end{array}$	$((\sim q \vee \sim s) \wedge (p \to q) \wedge (r \to s)) \to (\sim p \vee \sim r)$	Destructive Dilemma

Problem 1. Check whether the following argument is a valid or not.

$$\begin{array}{c} p \vee q \\ q \to r \\ p \to s \\ \sim s \\ \hline \therefore r \end{array}$$

Solution. Here the premises are $p \vee q$, $q \to r$, $p \to s$ and $\sim s$. The steps for deducing whether r is a valid conclusion from the above premises are as follows:

Steps	**Statements**	**Applied Rules**
1.	$p \to s$	Rule P
2.	$\sim s$	Rule P
3.	$\sim p$	Rule T, Step 1, Step 2 and I_5
4.	$p \vee q$	Rule P
5.	q	Rule T, Step 3, Step 4 and I_7
6.	$q \to r$	Rule P
7.	r	Rule T, Step 5, Step 6 and I_4

Thus, r is a valid conclusion and hence the given argument is valid.

Problem 2. Check whether the following argument is a valid or not.

$$\begin{array}{c} p \to (q \to s) \\ \sim r \vee p \\ q \\ \\ \hline \therefore r \to s \end{array}$$

Solution. Given premises are $p \to (q \to s)$, $\sim r \vee p$, and q. The steps for deducing whether $r \to s$ is a valid conclusion from the above premises are as follows.

Steps	**Statements**	**Applied Rules**
1.	$\sim r \vee p$	Rule P
2.	r	Rule P (assumed premise)
3.	p	Rule T, Step 1, Step 2 and I_7
4.	$p \rightarrow (q \rightarrow s)$	Rule P
5.	$q \rightarrow s$	Rule T, Step 3, Step 4 and I_4
6.	q	Rule P
7.	s	Rule T, Rule 5, Rule 6 and I_4
8.	$r \rightarrow s$	Rule CP

Thus, $r \rightarrow s$ is a valid conclusion from the given premises. Hence, the given argument is valid.